

Coding Theory / Codes & group codes

Word: sequence of 0's and 1's

Code: collection of words

weight: NO of 1's present in code

eg $x = 1101$
 $w(x) = 3$

Ex-OR operation

$$0 \oplus 0 = 0 \quad 1 \oplus 0 = 1$$

$$0 \oplus 1 = 1 \quad 1 \oplus 1 = 0$$

eg $x = 011$, $y = 001$

$$x \oplus y = 010$$

$$w(x \oplus y) = 1$$

The distance between x & y is the weight of $x \oplus y$ also called Hamming distance.

→ Encoding function

Encoding function is defined as

$e: B^m \rightarrow B^n$, where e is (m, n) encoding function

eg. $e: B^2 \rightarrow B^6$ e is $(2, 6)$ encoding function

→ Detection of errors

Let $e: B^m \rightarrow B^n$ is an encoding function & the minimum distance of e is $k+1$, then e can detect k or less errors.

→ Correction of errors

Let $e: B^m \rightarrow B^n$ is an encoding function & the minimum distance of e is $2k+1$ then it can correct k or less errors.

type 1: Detection and correction of errors

① Consider the $(2, 4)$ encoding function.
How many errors can e detect and how many errors can e correct

$$e(00) = 0000$$

$$e(10) = 0110$$

$$e(01) = 1011$$

$$e(11) = 1100$$

Ans

$$e: B^m \rightarrow B^n$$

As per question

$$e: B^2 \rightarrow B^4$$

$$\text{Let } X_0 = 0000$$

$$X_1 = 0110$$

$$X_2 = 1011$$

$$X_3 = 1100$$

$$X_0 \oplus X_1 = 0110$$

$$X_0 \oplus X_2 = 1011$$

$$X_0 \oplus X_3 = 1100$$

$$X_1 \oplus X_2 = 1101$$

$$X_1 \oplus X_3 = 1010$$

$$X_2 \oplus X_3 = 0111$$

$$w(X_0 \oplus X_1) = 2$$

$$w(X_0 \oplus X_3) = 2$$

$$w(X_1 \oplus X_3) = 2$$

$$w(X_0 \oplus X_2) = 3$$

$$w(X_1 \oplus X_2) = 3$$

$$w(X_2 \oplus X_3) = 3$$

minimum distance = 2

Detection of errors

$$k+1 = e$$

$$k+1 = 2$$

$$\Rightarrow k = 1$$

$\therefore$ so it can detect one or less errors

Correction of errors

$$2k+1 = e$$

$$\Rightarrow 2k+1 = 2$$

$$\Rightarrow k = 1/2$$

$\therefore$ can correct zero error.

type-2 Group code

problem show that $(2,5)$ encoding function defined by $e(00) = 00000$, $e(01) = 01110$, $e(10) = 10101$, $e(11) = 11011$ is a group code.

solⁿ closure property

$\oplus$	00000	01110	10101	11011
00000	00000	01110	10101	11011
01110	01110	00000	11011	10101
10101	10101	11011	00000	01110
11011	11011	10101	01110	00000

Group code encoding function $\oplus$
Ex - OR

All the entries of the table belongs to e .
 $\therefore e$ is closed

Associative

$$a * (b * c) = (a * b) * c$$

$$00000 \oplus (01110 \oplus 10101) = (00000 \oplus 01110) \oplus 10101$$

$$(00000) \oplus (11011) = (01110) \oplus 10101$$

$$11011 = 11011$$

$\therefore$ Associative property is satisfied

Identity element

$$a * e = a$$

$$00000 \oplus 00000 = 00000$$

$$01110 \oplus 00000 = 01110$$

$$10101 \oplus 00000 = 10101$$

$$11011 \oplus 00000 = 11011$$

Inverse

$$a * a^{-1} = e$$

$$00000 \oplus 00000 = 00000$$

$$01110 \oplus 01110 = 00000$$

$$10101 \oplus 10101 = 00000$$

$$11011 \oplus 11011 = 00000$$

Inverse exist

Commutative

$$a * b = b * a$$

or diagonal opposite entries are same

$\therefore$ encoding function $e: B^2 \rightarrow B^5$ is a group

type-3: encoding function

parity check matrix

$$H = \begin{bmatrix} \text{identity matrix} \end{bmatrix} \begin{matrix} m \text{ row} \\ n \text{ row} \end{matrix}$$

$$H = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} m=2 \\ n=5 \end{matrix}$$

① consider the parity check

determine the group code
encoding function.

$$\rightarrow e: B^m \rightarrow B^n$$

$$e: B^2 \rightarrow B^5$$

$$e(00) = ?$$

$$e(01) = ?$$

$$e(10) = ?$$

$$e(11) = ?$$

$$e(00) = 00x_1x_2x_3 \text{ mod 2 Boolean pr}$$
$$[x_1x_2x_3] = [00] \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$00000 = [0 \oplus 0 \quad 0 \oplus 0 \quad 0 \oplus 0]$$
$$= [000]$$

$$e(00) = 00000$$

$$e(01) = 01x_1x_2x_3$$

$$[x_1x_2x_3] = [01] \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = [0 \oplus 0 \quad 0 \oplus 1 \quad 0 \oplus 1]$$

$$[x_1x_2x_3] = [011]$$

$$e(01) = 01011$$

$$e(10) = 10x_1x_2x_3$$

$$[x_1x_2x_3] = [10] \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = [1 \oplus 0 \quad 1 \oplus 0 \quad 0 \oplus 0]$$

$$[x_1x_2x_3] = [110]$$

$$e(10) = 10110$$

$$e(11) = 11x_1x_2x_3$$

$$[x_1x_2x_3] = [11] \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = [1 \oplus 0 \quad 1 \oplus 1 \quad 0 \oplus 1]$$

$$[x_1x_2x_3] = [101]$$

$$e(11) = 11101$$

then draw the composition table to verify the group code.

type-4 : Decoding function [maximum likelihood test] Teaching

Q Consider $(2,5)$ group encoding function

$e: B^2 \rightarrow B^5$ defined by as

$$e(00) = 00000 \quad e(01) = 01110$$

$$e(10) = 10101 \quad e(11) = 11011$$

Decode the following words relative to maximum likelihood function

$$(i) \quad 11110$$

$$(ii) \quad 10011$$

$$(iii) \quad 10100$$

Sol':

00000	01110	10101	11011
00001	01111	10100	11010
00010	01100	10111	11001
00100	01010	10001	11111
01000	00110	11101	10011
10000	11110	00101	01011

$$e(01) = 11110$$

$$e(10) = 10100$$

$$e(11) = 10011$$

type-4 problem-2: Decoding function [maximum likelihood]

② Let $H = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \left\{ \begin{array}{l} m=3 \\ \rightarrow n=6 \end{array} \right.$

Decode the following words

① 011001 (i) 101001 (ii) 111010

$$\Rightarrow e: B^3 \rightarrow B^6$$

$$e(000) = 000x_1x_2x_3$$

$$[x_1x_2x_3] = [000] \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = [000]$$

$$e(000) = 000000$$

$$e(001) = 001x_1x_2x_3$$

$$[x_1x_2x_3] = [001] \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = [011]$$

$$e(001) = 001011$$

$$e(010) = 010110$$

$$e(100) = 100100$$

$$e(011) = 011101$$

$$e(110) = 110010$$

$$e(101) = 101111$$

$$e(111) = 111001$$

000000	001011	010110	011101	100100	101111	110010	111001
000001	001010	010111	011100	100101	101110	110011	111000
000010	001001	010100	011111	100110	101101	110000	111010
000100	001111	010010	011001	100000	101011	110110	111101
001000	000011	011110	010101	101100	100111	110101	110001
010000	011011	000110	001101	110100	111111	100010	101001

$$\checkmark e(011) = 011001$$

$$\checkmark e(110) = 110010$$

$$\checkmark e(111) = 101001$$

Type-3: Encoding function

② Let $H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ be a parity check matrix

Determine the group code.

$$\Rightarrow e: B^m \rightarrow B^n$$

$$e: B^3 \rightarrow B^6$$

$$e(000) = ? \quad e(011) = ?$$

$$e(001) = ? \quad e(101) = ?$$

$$e(010) = ? \quad e(110) = ?$$

$$e(100) = ? \quad e(111) = ?$$

$$e(000) = 000x_1x_2x_3$$

$$[x_1x_2x_3] = [0 \ 0 \ 0] \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= [0 \oplus 0 \oplus 0 \ 0 \oplus 0 \oplus 0 \ 0 \oplus 0 \oplus 0]$$

$$= [0 \ 0 \ 0]$$

$$\boxed{e(000) = 000000}$$

$$e(001) = 001x_1x_2x_3$$

$$[x_1x_2x_3] = [0 \ 0 \ 1] \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= [0 \oplus 0 \oplus 1 \ 0 \oplus 0 \oplus 1 \ 0 \oplus 0 \oplus 1]$$

$$= [1 \ 1 \ 1]$$

$$\boxed{e(001) = 001111}$$

$e(010) = 010011$	$e(101) = 101011$
$e(100) = 100100$	$e(110) = 110111$
$e(111) = 111000$	$e(011) = ? (H0)$

Next verify the generated code by drawing comparison table.